higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – SHEET METAL WORK
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
- It provides insulation.
- It provides soundproofing. (2 × 1) (2)

1.2 A seam is where two pieces of metal are joined. (2)

1.3
- To avoid severe throttling of the damper, select all terminals for a zone or room at approximately the same pressure drop.
- To prevent the creation of draughts, ensure that no airflow from a terminal exceeds the distance to a sidewall.
- To prevent the creation of stagnant pockets, ensure even distribution of air in a room.
- Ensure that ceiling or wall terminals do not discharge directly on a column or any other permanent obstruction. (4)

1.4 *Acoustic acquisition for ductwork* refers to the control of noise in ductwork. (2)

1.5
- Fans and their motors and drives
- Air filters
- Coils and drip trays
- Air dampers and linkages
- Chilled water
- Hot water and refrigeration piping and their control devices
- Drain and overflow piping (Any 5 × 1) (5)

[15]

QUESTION 2

2.1
2.1.1 True
2.1.2 True
2.1.3 False
(3 × 1) (3)

2.2
- Draw all lines uniform and bold in black.
- Draw all lines so that they have the same density.
- Draw all lines for one drawing either in pencil or ink.
- Draw all lines with the correct thickness. (4 × 1) (4)

2.3 A grill is a perforated or louvred✓ covering on any area✓ where air passes through.✓ (3)

2.4
- Different connectors
- Rectangle elbows
- Square elbows
- Transition elbows
- Offsets
- Transition pieces
- Round tees
- Round elbows
- Diffusers
- Grilles
- Air control units
- Any other equipment, including fans and terminal units, that is included in the duct design (Any 5 × 1) (5)

[15]

QUESTION 3

3.1
- Gas
- Tungsten
- Arc
- Welding (Any 2 × 1) (2)

3.2 A – Filler rod
B – Electrode
C – Cap
D – Earth clamp
E – Weld bead (5 × 1) (5)

3.3 The primary purpose of shielding gas is to prevent exposure of the molten weld pool to oxygen, nitrogen and hydrogen contained in the atmosphere. (3)

3.4 No (1)

3.5 3.5.1 B
3.5.2 C
3.5.3 D
3.5.4 A

(4)

[15]

QUESTION 4

4.1
- Turn on the equipment and set the correct welding amperage on the machine.
- Open the shielding gas cylinder and adjust the gas flow to the correct setting.
- Now strike the arc by quickly touching the electrode to the work and rapidly lifting it off the surface by a short distance. (3 × 1) (3)

4.2
- The cup or nozzle
- The weld pool size
- Air movement (3 × 1) (3)

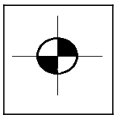
4.3
- Accident protection
- Enough ventilation
- Proper lighting
- Fire protection
- The levels of cleanliness
- First aid in the workplace
- The use of electric and pneumatic power (7 × 1) (7)

4.4 *Cathode* is the negatively charged terminal✓ in an electric circuit.✓ (2)
[15]

QUESTION 5

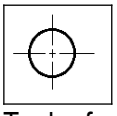
5.1
- Reading data
- Converting data into co-ordinates for the machine.
- Checking the machine limitations like allowable feeds and speeds.
- Developing suitable speeds and feed for the machine.
- Developing motion instructions for the machine.
- Setting codes for various instructions.
- Printing out a listing of the programs for checking.
- Printing out error codes. (8 × 1) (8)

5.2 5.2.1



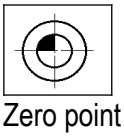
Reference point (R)

5.2.2



Tool reference point (N)

5.2.3



Zero point (M)

(2 × 3) (6)

5.3 5.3.1 The programme that tells the computer how to do a particular job

5.3.2 The equipment and devices that make up a computer system

5.3.3 Set of instructions used to solve mathematical problems

5.3.4 A two-dimensional or three-dimensional toolpath generated by the programme codes

(4 × 2) (8)

5.4
- The chuck
- The cutting tool
- Tail stock
- Bed

(4 × 1) (4)

5.5 The editing procedure will depend on the type of_controller✓ the machine has by using the machine codes✓ to do the editing. Editing will happen only after you have tested the program✓ or made the first component.✓ (4)

5.6 5.6.1 C & I
5.6.2 E & G
5.6.3 F & J
5.6.4 A & D
5.6.5 B & H

(2 × 5) (10)
[40]

TOTAL: 100

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